Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #4

Posted: Monday, Feb 23 **Due: Tuesday, Mar 3, 11:59 pm**

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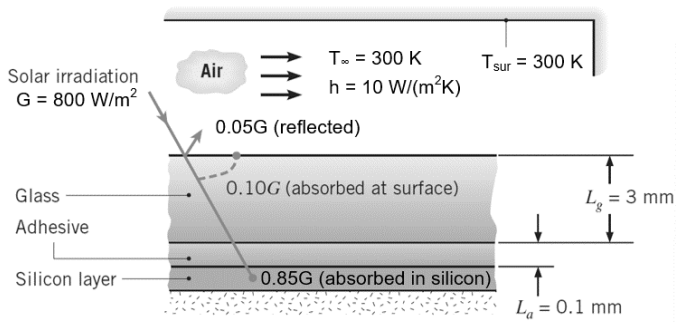
1. (40 pt) A photovoltaic (PV) panel consists of (top to bottom) a 3-mm-thick ceria-doped glass (k_g = 1.5 W/m·K), a 0.1-mm-thick optical grade adhesive (k_a = 1 W/m·K), and a very thin layer of silicon within which solar energy is converted to electricity. The solar-to-electricity conversion efficiency within the silicon layer η decreases with increasing silicon temperature T_{Si}, as described by the expression $\eta = a - bT_{Si}$, where a = 0.553 and b = 0.001 K^{-1}. The temperature T_{Si} is expressed in Kelvins. The backside of the silicon layer is thermally insulated. Contact thermal resistance between the adhesive and silicon layer is $R_c = 10^{-4}$ m^2·K/W. There is no contact resistance between the glass and the adhesive.



In a sunny day, there is G = 800 W/m^2 solar irradiation reaching the PV panel. 5% of solar irradiation is reflected from the top surface of the glass, 10% of solar irradiation is absorbed at the top surface of the glass, and 85% of solar irradiation is transmitted to and absorbed within the silicon layer. Part of the solar irradiation absorbed in the silicon is converted to thermal energy and the remainder is converted to electricity. The PV panel is exposed to ambient air with temperature of T_∞ = 300 K and convective heat transfer coefficient of h = 10 W/m^2·K. The surrounding temperature is $T_{sur} = T_\infty$ = 300 K and the effective radiative heat transfer coefficient is h_{rad} = 5 W/m^2·K. Hint: you do not need to use the Stefan-Boltzmann law to calculate the radiative heat loss.

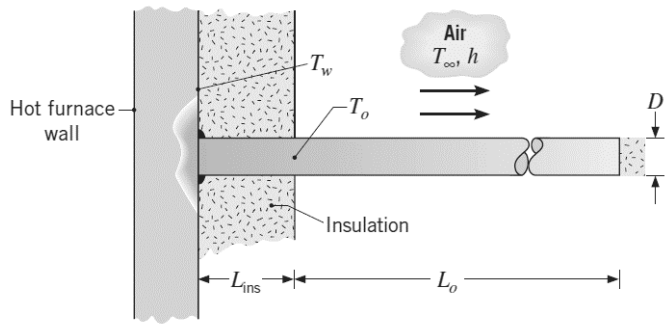
(a) Show the thermal resistance network that can describe heat transfer across the PV panel. Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the

temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network.

(b) Calculate the temperatures of the top surface of the glass $T_{g,top}$ and the silicon layer T_{Si}.

(c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

2. (30 pt) A rod of diameter D = 10 mm and thermal conductivity k = 80 W/m·K protrudes normally from a furnace wall that is at T_w = 200 °C and is covered by insulation of thickness L_{ins} = 200 mm. The rod is welded to the furnace wall and is used as a hanger for supporting instrumentation cables. The rod has an exposed length L_o and its tip is well insulated. To avoid damaging the cables, the temperature of the rod at its exposed surface, T_o, must be maintained below a specified operating limit of T_{max} = 100 °C. The ambient air temperature is T_∞ = 20 °C and convective heat transfer coefficient is h = 20 W/m^2·K.



(a) Derive an expression for the exposed surface temperature T_o as a function of the prescribed thermal geometrical parameters.

(b) If exposed length L_o is 200 mm, calculate the corresponding exposed surface temperature T_o. Does the exposed length meet the specified operating limit?

(c) Calculate the minimum exposed length L_o that ensures T_o below the specified operating limit.

3. (30 pt) A square pin fin of uniform cross-section area is fabricated of copper (k = 400 W/m·K). The fin width is s = 10 mm and the fin is exposed to air flow with convective heat transfer coefficient h = 100 W/m^2·K. It is known that the fin efficiency is η_f = 0.6. The fin has a convection tip.

(a) Calculate the fin length L.

(b) Calculate the thermal resistance of the fin R_f and the fin effectiveness ε_f.

1.) Given

Radiation:	Convection	conduction
$G = 800 W/m^2$	$T = 300°K$	$L_g = 3mm$ $k_g = 1.5 W/m°K$
$400 W/m^2$ reflect	$T_{sky} = 300°K$	$L_A = 0.1mm$ $k_A = 1 W/m°K$
400W absorbed by glass & 80 absorbed by silicon	$h = 10 W/m^2°K$	
	$\tau = 0.1mm$ (thickness)	

Other things

$\eta = 0.533 - 0.001 K^{-1} T_{si}$ $R_e = 10^{-4} m^2°K$

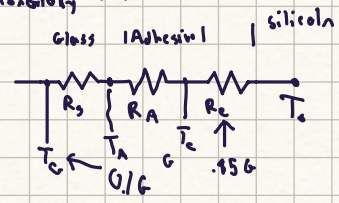
c.) $\eta = 0.0533 \cdot \frac{342.8}{1000} = 0.2107$

$q'' = \eta \cdot 660 = 143.34$

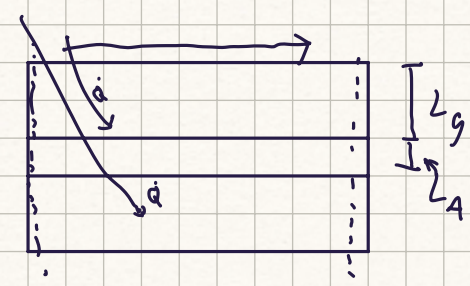
Assumptions: 1-D steady state

silicon remains flat

Schematic



Radiation



$R_g = \frac{L_g}{k_g} = 2 \cdot 10^{-3}$ $R_A = \frac{1}{1} = 0.1 \cdot 10^{-3}$

$R_{cond} = R_g + R_A + R_e = 0.002$

$R_{ext2} = \frac{1}{h_{ext} \text{ conv}} = \frac{1}{15}$

$R_{ext} = 0.06$

$\frac{T_{c\,top} - T_\infty}{R''_{Ext}} = q''_1 + 80$

$q''_1 = \frac{T_{si} - T_{g,top}}{R_{cond}}$

We can set the equal

Plugging into wolfram alpha we find

$T_{si} = 342.8$

$T_{g,top} = 341$

2.)

Given: $D = 0.01m$ $k = 80 W/m\cdot K$ $T_w = 200°C$ $L_{ins} = L = .2m$

Insulate to $T_{max} = 100°C$ $T_\infty = 20°C$ $h = 20 W/m^2 K$

schematic:

Find: T_0 (a) make an expression for T_0

$A_c = \pi R^2$ $P = 2\pi R$ $R = D/2$

$m = \sqrt{\frac{hP}{kA_c}}$

$\frac{T_w - T_0}{L} = m(T_0 - T_\infty)\tanh(mL_0)$

$T_0(L_0) = \frac{T_w + L_{ins} m \tanh(mL_0) T_\infty}{1 + L_{ins} m \tanh(mL_0)}$

B.) Given $L_0 = 200mm$

find T_0

plug into $T_0 = \frac{200 + .2 \tanh(18.2)20°C}{1 + .2 \tanh(10 \cdot .2)}$

$T_0 = 81.5mm$, yes it meets the operating limits.

C.) Given $T_{max} = 100°C$

find L_0

$100 = \frac{200 + .2 \tanh(164) \cdot 20°C}{1 + \tanh(10x)} = .073 = \underline{73mm}$

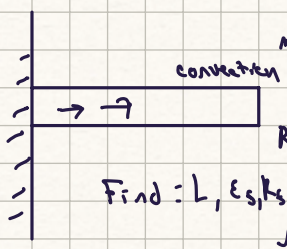
Discussion: From A→C we've optimized the lengths of the fin

3.) Given: $k = 400\ W/m\cdot{}^\circ K$ $h = 100\ W/m^2{}^\circ K$

$s = 16\ mm$ $\theta_B = 16\ m$

$\eta_f = 0.6$

schematic



$m = \sqrt{hp}$

Assumptions: Convective fin
standard mode

$R_s = \frac{\theta_B}{q_f} =$

Find: L, ε_s, R_s

$\eta = \frac{\tanh(mL_c)}{mL_c}$ $\sqrt{\frac{hp}{kA_c}}$

p = perimeter

$p = 4s = 0.004$

$m = \sqrt{\frac{hp}{kA_c}} = 10\ \frac{1}{m}$

$L = L_c + \frac{A_c}{p}$

$\eta_f = \frac{\tanh(mL_c)}{mL_c}$

$.6 = \frac{\tanh(10L_c)}{10L_c}$

$L_c = 0.15122$

$L = 0.149\ m$

B.)

$R_t = \frac{1}{hA_s\eta_s}$

$\frac{m\tanh(mL)}{hA_c\theta_B}$

Discussion: All our values seem reasonably physical & are correct

Summary
A

$A_s = PL + A_c$

$R_t = \frac{1}{\eta_s h A_s}$ $0.6 \cdot 100\,(PL + A_c)$

$R_t = 2.57\ K/W$

$\varepsilon_s = \frac{A_f}{A_c} = 0.00604 \cdot 10^4 \cdot 0.6$

$= 36.3$